PROJECT:
Network Design for Faculty of Computing Block N28B

TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

SECR1213 - NETWORK COMMUNICATIONS
SEMESTER I, SESSION 2024/2025
SECTION 06

LECTURER: DR. RAJA ZAHILAH BINTI RAHA MOHD. RADZI

GROUP NAME: NETWORK NEXUS

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TASK 1: PROJECT SETUP

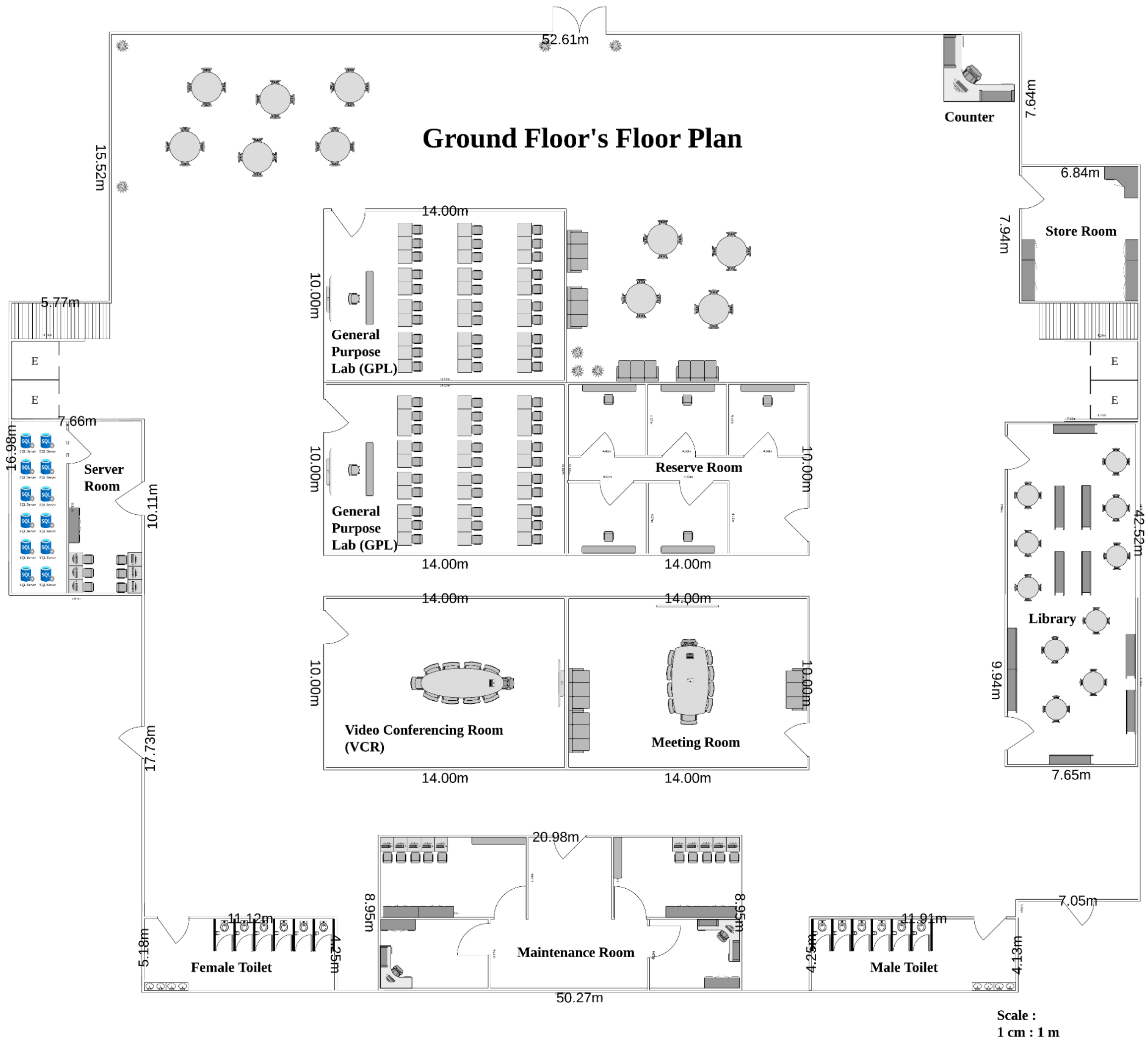
1.0 Introduction

In Task 1 of the project, our group, "Network Nexus," was tasked with designing a floor plan for the new 2-storey building, N28B, for the Faculty of Computing (FC). This building is being developed to support the anticipated 15% growth in both student and academic staff populations over the next four years. The new facility will include essential spaces such as four labs, a video conferencing room, a hybrid classroom, and a student lounge with Wi-Fi connectivity.

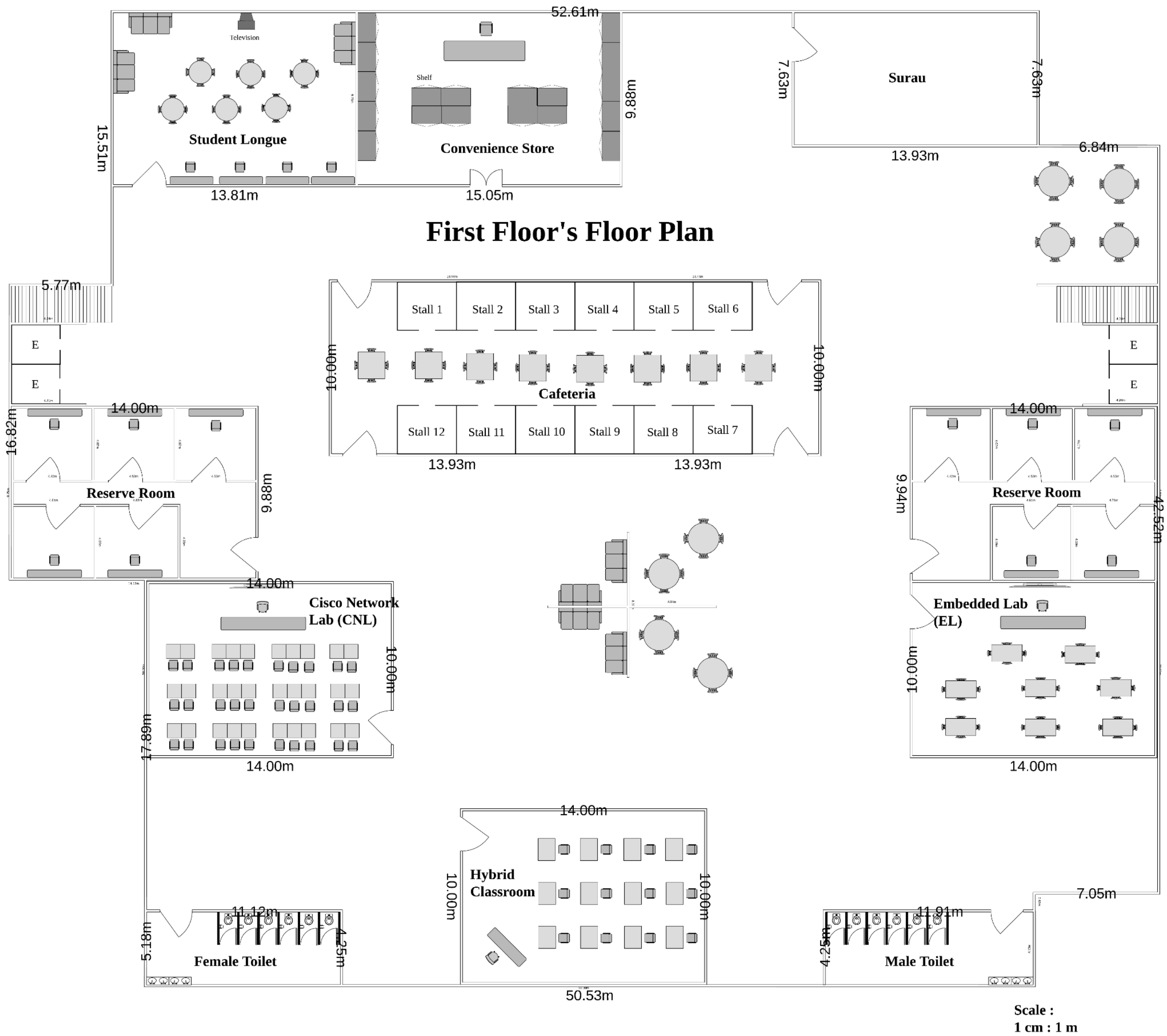
The design must align with the Faculty's goal of creating a scalable, cost-effective, and future-ready environment. It will support high-speed internet connectivity, reliable network security, and future wireless capabilities, all while ensuring efficient use of space and resources. Our team has developed a floor plan that meets these requirements and supports the FC's vision for future growth and technological advancement.

2.0 Floor Plan Design

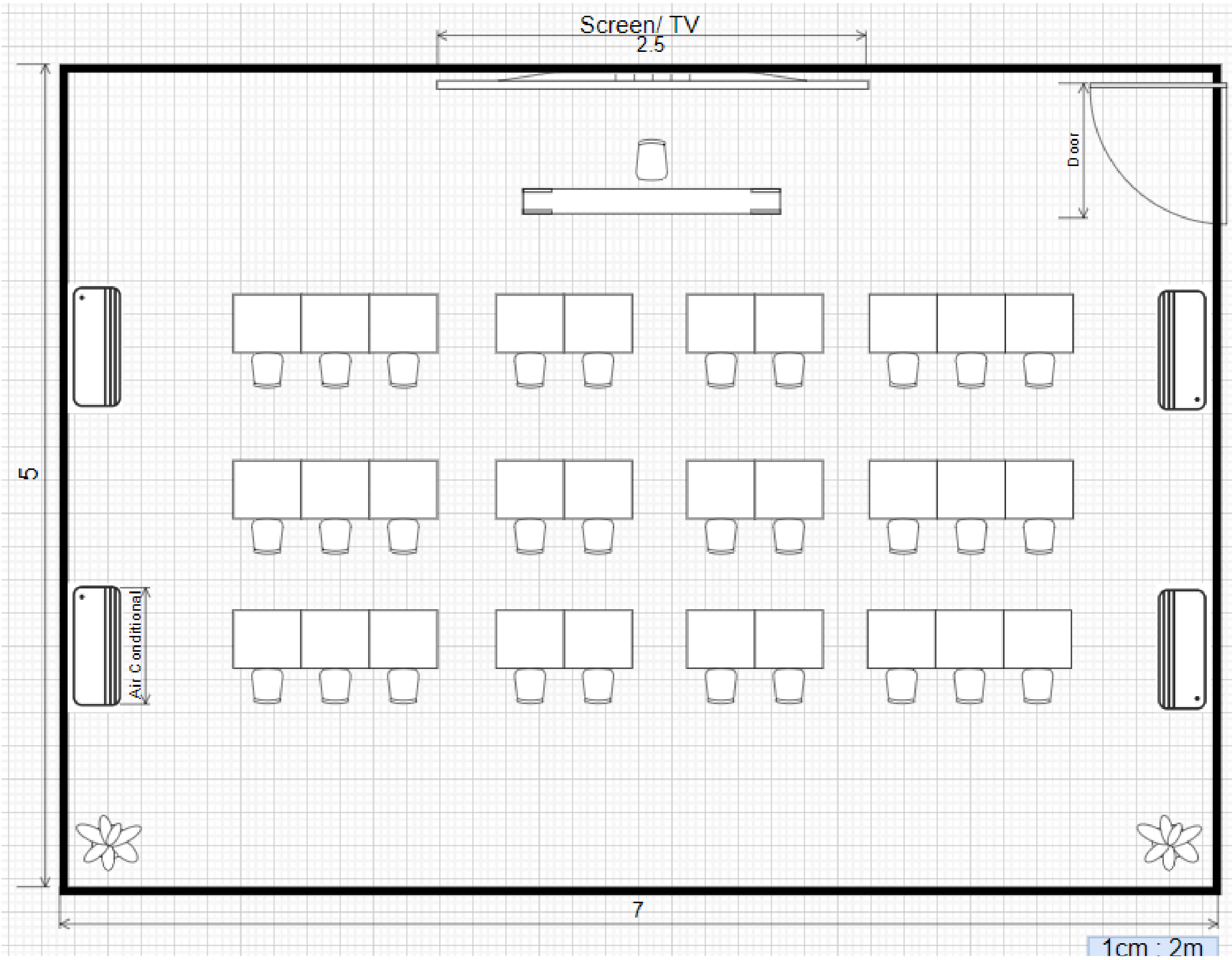
2.1 Ground Floor's Floor Plan



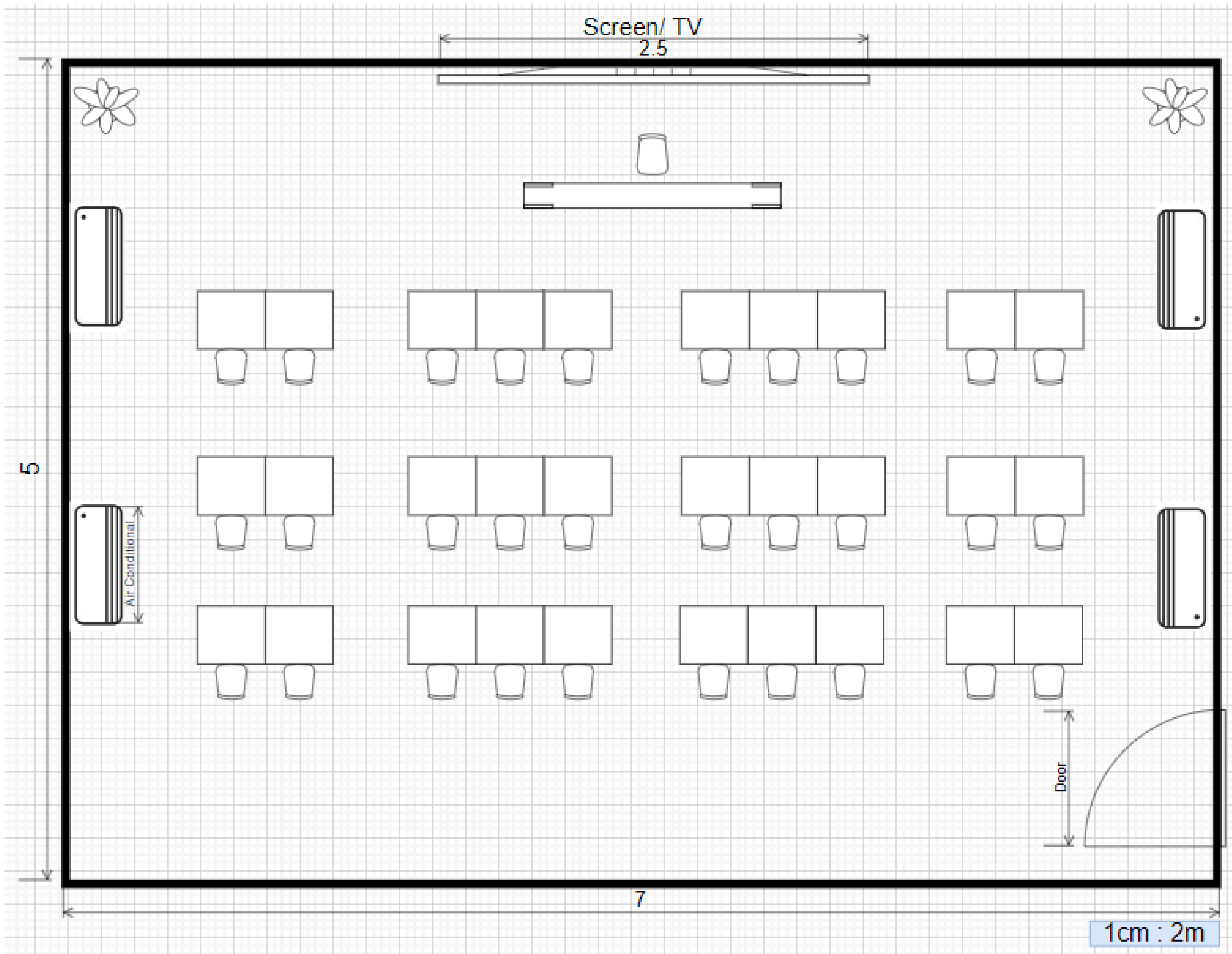
2.2 First Floor's Floor Plan



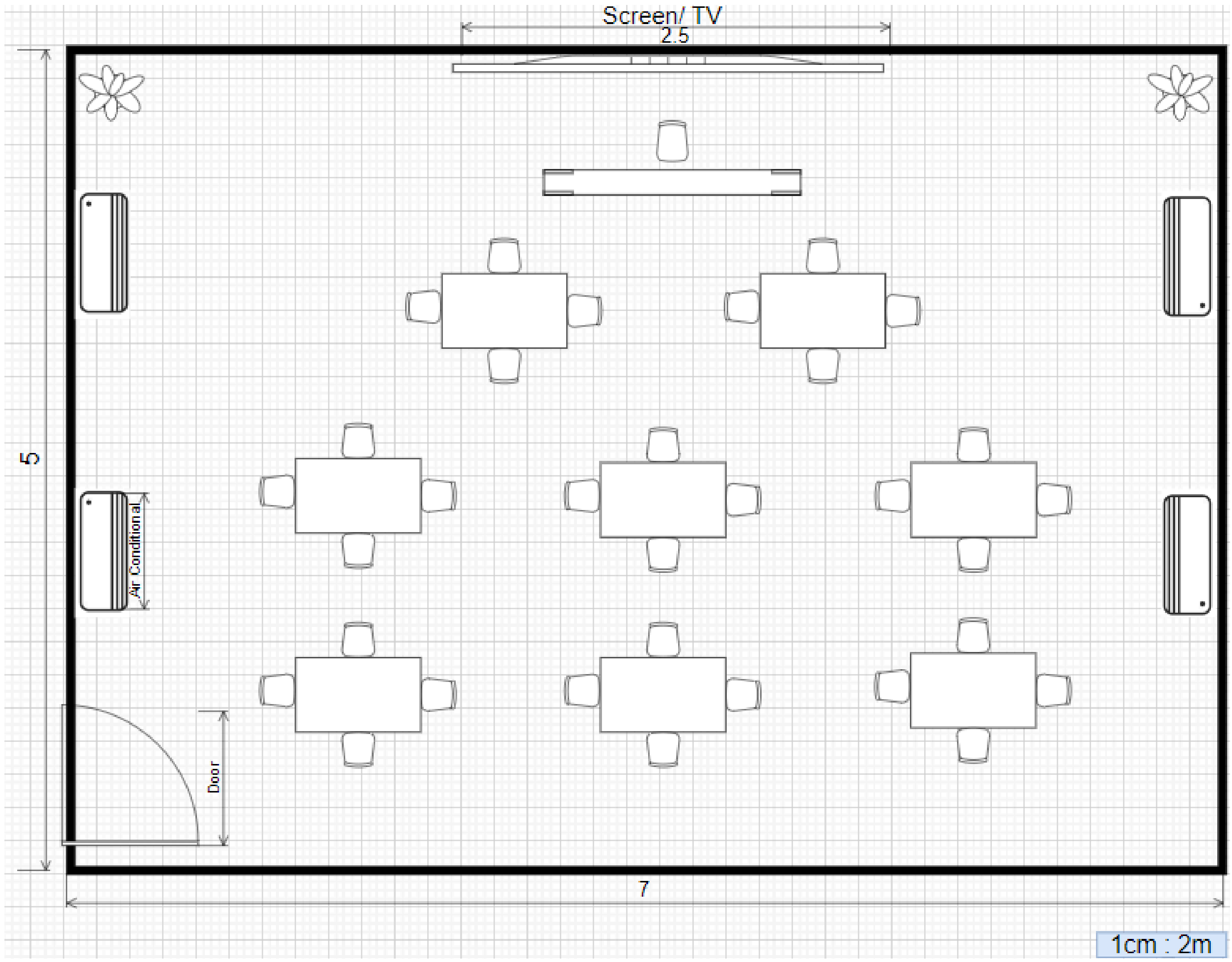
2.3 Floor Plan for General Purpose Lab



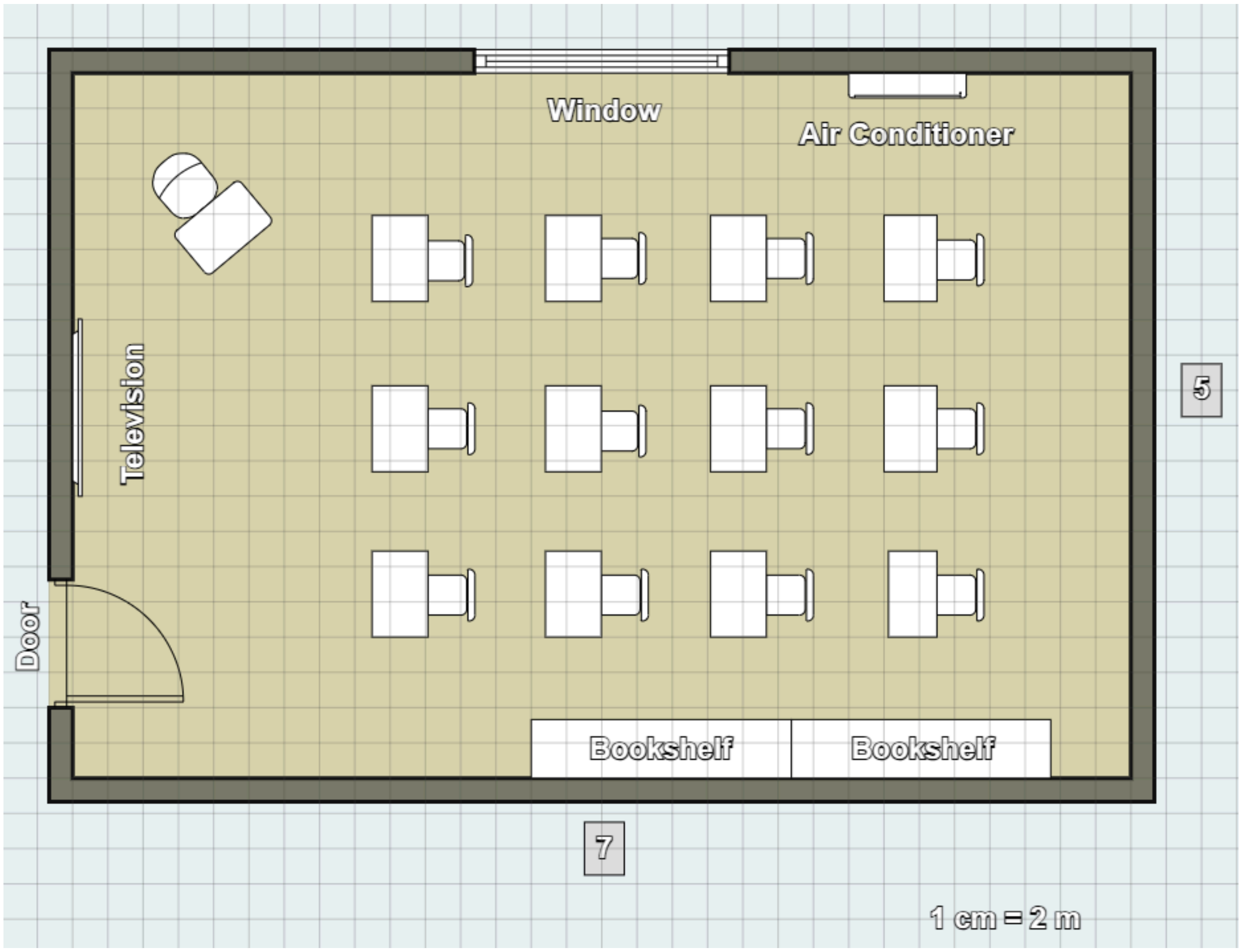
2.4 Floor Plan for Cisco Network Lab



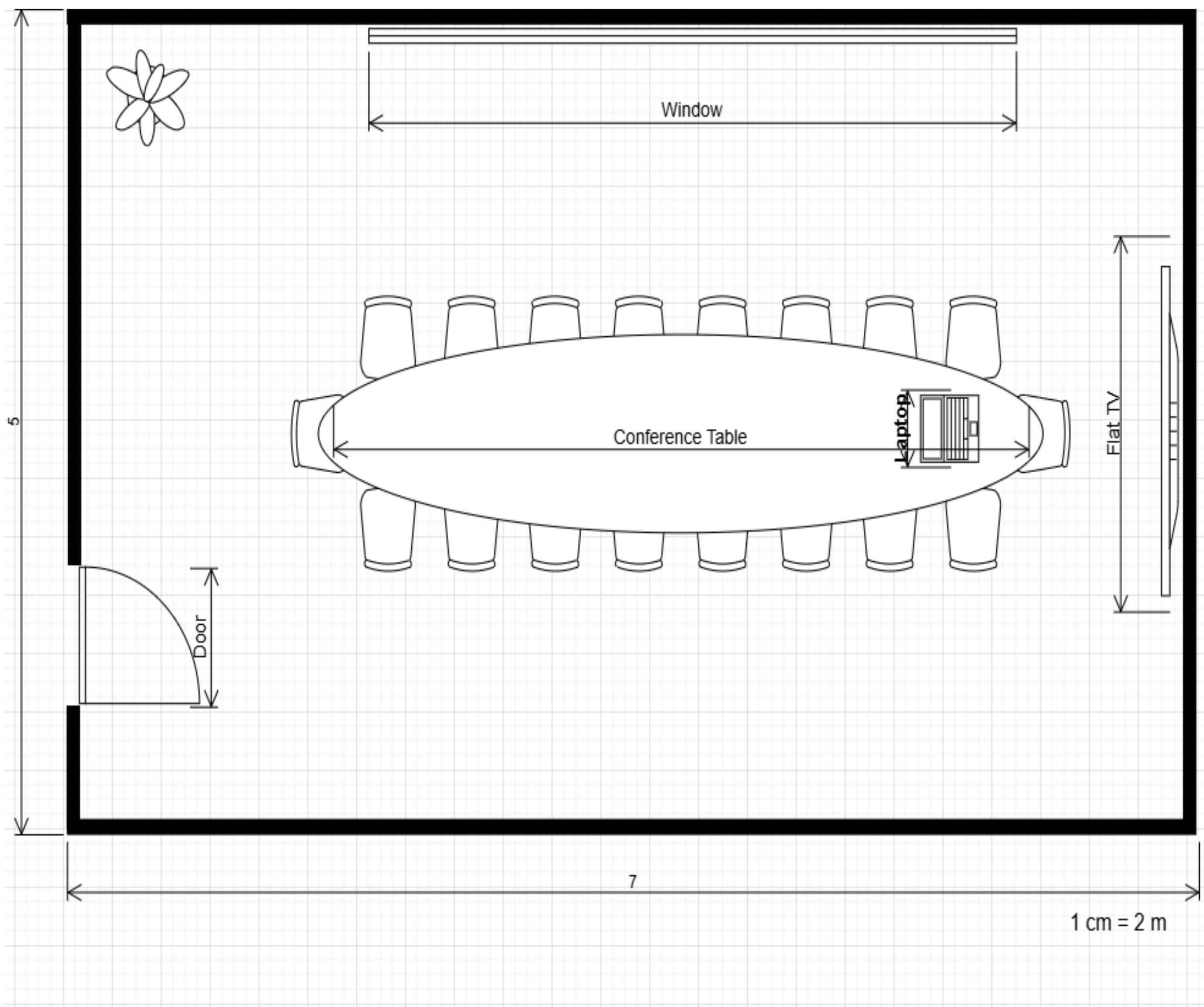
2.5 Floor Plan for Embedded Lab (IOT, Digital and sensors)



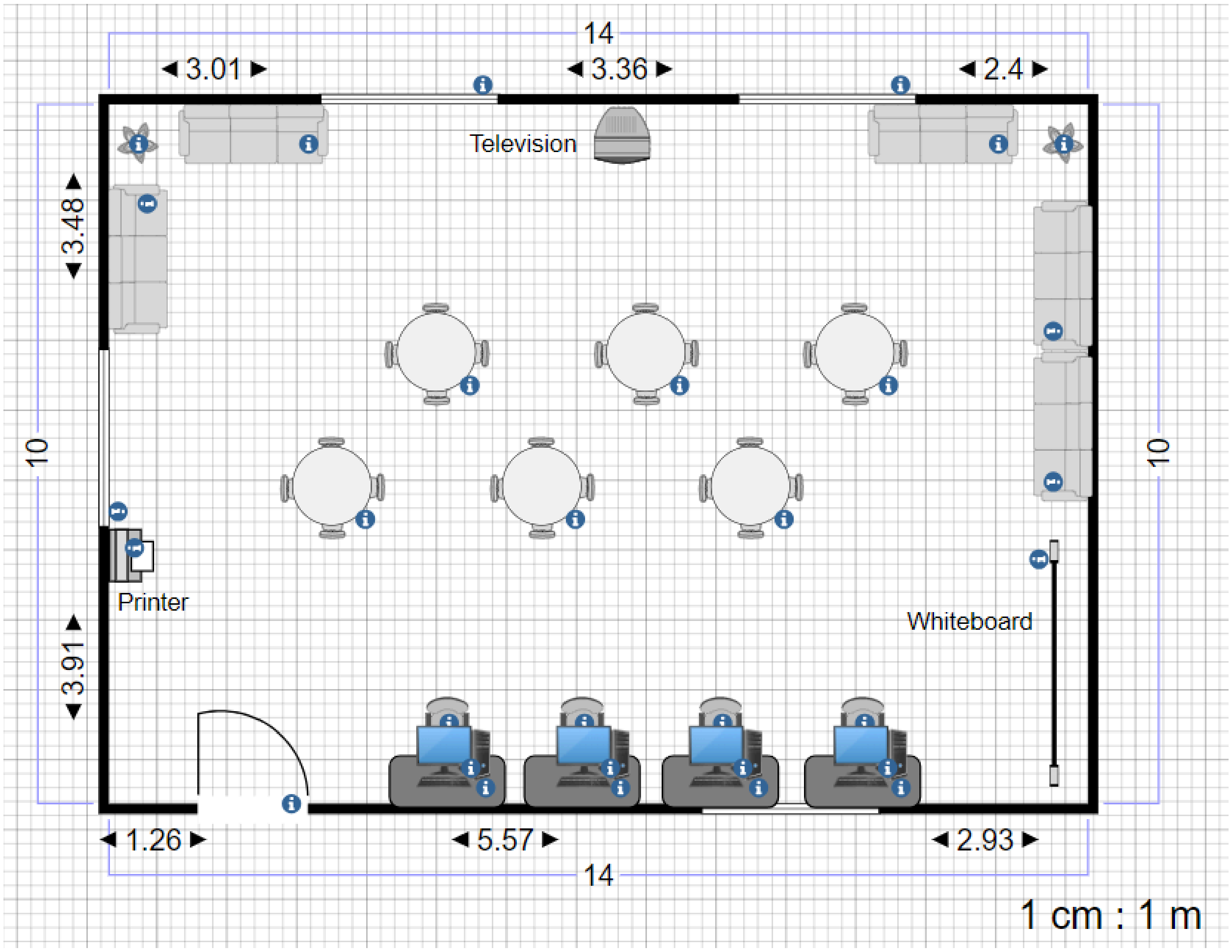
2.6 Floor Plan for Hybrid Classroom



2.7 Floor Plan for Video Conferencing Room

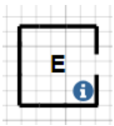


2.8 Floor Plan for Student Lounge

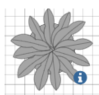


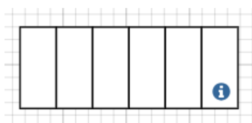
2.9 Symbol List

GPL - General Purpose Lab
VCR - Video Conferencing Room
CNL - Cisco Network Lab
EL - Embedded Lab

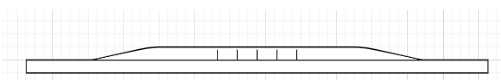


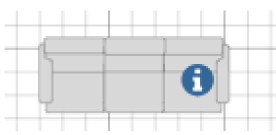
- Elevator

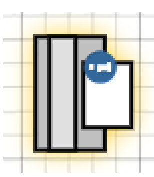
 - Plant

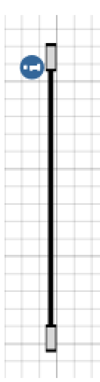
 - Stair

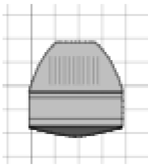
 - Air Conditioner

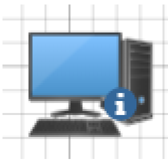
 - TV/Screen

 - Sofa

 - Printer

 - Whiteboard

 - Television

 - Computer / PC

TASK 2: INITIAL DESIGN - PRELIMINARY ANALYSIS

3.0 Introduction

In task 2, the project from the setup part moves on to the initial design. This stage is for us to do preliminary analysis. Which is focusing on understanding and designing the network requirements for the Faculty of Computing's new block N28B building. This phase centers on assessing the infrastructure, security measures, and unique technological needs across different areas in the building, including labs, classrooms, and the student lounge. By addressing key questions and conducting thorough research, the analysis explores vital components such as network infrastructure, required bandwidth, data protection, and monitoring systems. This will help ensure the network design is practical, reliable, and equipped to handle future demands. Ultimately, this preliminary review aims to confirm the technical viability of the project, aligning it with the Faculty's goals for a secure, scalable, and high-performance network.

4.0 Questions for understanding requirement of project from 3 dimension

4.1 Service

1. What kind of backup solutions should be implemented to ensure data protection and recovery in case of system failures?

   We propose implementing a hybrid backup system that combines both on-premises backups (local storage) and cloud-based backups, leveraging services like UTM's cloud infrastructure or external platforms such as AWS, Google Cloud, or Microsoft Azure. This approach ensures fast recovery from local backups for immediate needs, while maintaining off-site redundancy for disaster recovery, safeguarding critical data in case of catastrophic failure [1].

2. What kind of network monitoring and management tools should be implemented to ensure the smooth operation of the network?

   Based on expert recommendations, we propose implementing the SolarWinds Network Performance Monitor (NPM) for comprehensive network monitoring. This tool offers in-depth insights into traffic, bandwidth usage, and device performance, allowing real-time detection of network bottlenecks and outages [2]. With SolarWinds NPM, network administrators can act immediately to resolve issues, ensuring smooth and continuous operations.
   By utilizing this robust monitoring tool, SolarWinds NPM, we can ensure the efficient and reliable operation of the network infrastructure in building N28B, minimizing downtime and optimizing performance.

3. What type of firewall and security services should be integrated to prevent unauthorized access and protect sensitive data?

   To ensure robust network security, we recommend integrating Next-Generation Firewalls (NGFWs), which provide advanced security features such as deep packet inspection, intrusion prevention, and malware protection [3]. Given the Cisco Network Lab setup in the case study, implementing Cisco Firepower would be an ideal solution. Cisco Firepower integrates seamlessly with the existing Cisco infrastructure, offering advanced threat protection, application visibility, and URL filtering capabilities.

   For secure remote access, a VPN UTM solution should be the primary choice. VPN UTM ensures secure, encrypted communication for faculty, staff, and students accessing the university's network from off-campus locations, protecting sensitive data and maintaining privacy.

In addition to firewalls and VPNs, anti-virus software is essential to protect the network from cyber threats. Based on UTM’s current infrastructure, deploying the anti-virus solution recommended by UTM Digital across all systems will provide protection against viruses, malware, and ransomware. Installing this anti-virus software on all lab workstations is crucial to ensure that no viruses or Trojan horse malware are present within the network.

By leveraging Cisco Firepower, VPN UTM, and UTM's existing anti-virus solution, the university can implement a cohesive and secure network strategy that provides comprehensive protection for both on-campus and remote users.

4. How can network maintenance be efficiently managed for each lab, considering the different usage requirements and equipment types?

   Through research, we propose using network segmentation with VLANs (Virtual Local Area Networks) to separate different labs. This approach allows maintenance tasks to be performed on specific sections of the network without disrupting other parts, enhancing operational efficiency [5][6].

   Additionally, remote management and diagnostic tools can be utilized to enable the maintenance team to troubleshoot and resolve network issues without needing to be physically present in the labs, significantly reducing downtime and ensuring faster response times during network incidents [7].

4.2 Faculty

5. What is the internet bandwidth provided in the faculty?

   The internet bandwidth provided in a faculty normally will depend on the size of the user base such as students, academic staff and administrative personnel, as well as the usage requirements for teaching, research and administrative operations. Some of the faculty will use the high demand networking such as Faculty of Computing or Faculty of Engineering, both of the faculty normally require greater bandwidth to support high-performance applications, online lectures and research activities that involve large datasets or computational tasks. In UTM, university offers faculty-level bandwidth ranging from 1 Gbps to 10 Gbps.

6. What are the plans for faculty expansion that would need additional network resources such as the construction of new buildings or rooms?

   The plans for faculty expansion that would need additional network resources such as the construction of new buildings or rooms may include new facilities requiring dedicated networks, enhanced WI-FI coverage and security protocols. UTM may increase the demands of the network resources and improve the digital facilities. Additional rooms of the building for the faculty will require network access, network switches, router and upgrading the cabling to ensure the network bandwidth can support the use of internet in the faculty.

7. What devices and technologies are currently used in the faculty, what if the faculty plan to upgrade or add any new device in the near future?

   The devices and technologies currently used in the faculty are desktop computers, laptops, projectors, smartboard in smart classroom, there is a facility for VR equipment and lab machinery that needs to connect to the internet. If the faculty plan to upgrade or add any new device, such as IOT-enabled devices or advanced computing system, then it will require a more advanced or high performance network as the requirement to do such upgrading. Not only that, the faculty IT department needs to do the survey of the specific network bandwidth allowance or network configurations.

8. What is the faculty's annual budget located for network infrastructure such as hardware and software?

   The faculty's annual budget is mostly consumption for network infrastructure, routine maintenance for hardware such as routers, switches and cabling, for software is software licenses and potential upgrade for the system. The amount of budget followed by the improvement in network bandwidth, hardware and cybersecurity. A secure and powerful system enabling UTM to stay at the forefront of digital learning.

4.3 Network

9. What is the expected performance for each lab, classroom and the student lounge? Are there specific activities that require higher network performance?

   Each lab, classroom, and lounge will require high-speed internet, because the performance needs are high due to practical experiments and software used. The hybrid classroom should support smooth video streaming and real-time collaboration tools.

Wi-Fi network coverage must be thorough and reachable throughout the new building, while focusing on high-needed areas like labs, hybrid classrooms, and the student lounge.

10. What are the requirements for the video conferencing room? What is the use of the room?

    The basic requirements of a video conferencing room are sufficient network bandwidth and high-resolution camera. The best video conferencing camera system, microphone and speaker are also essential and needed. The room will be used for virtual project meetings. It should support High Definition (HD) video, multi-screen displays, and cameras that allow multiple participants to be seen clearly. High-quality audio and video are also important, along with support for various video conferencing platforms.

11. What technologies are needed for the hybrid classroom to ensure effective teaching? Should it support both in-person and online learning simultaneously, and if so, what tools and devices are necessary?

    To ensure effective teaching, the hybrid classroom should be equipped with interactive displays and tools such as projector and smart whiteboard, audio and visual equipment such as speakers and high-quality cameras, and high-speed internet. Optimizing the arrangement of devices and tools is essential to ensure that both in-person and online students can learn more effectively. Yes, it should support both in-person and online learning with features like screen sharing, and real-time interaction between in-person and online students.

12. What are the detailed requirements for each lab in the new building? Are there specific educational tools or software that must be supported?

    The Cisco Network Lab requires Cisco networking devices, such as Cisco routers and Cisco switches. The Embedded Lab needs IoT kits, and various sensors and peripherals. Meanwhile the general-purpose labs should have modern workstations with high-performance specs for software development and simulations.

5.0 Feasibility of project

5.1 Legal Feasibility

After conducting research on the project based on legal aspects, our project is feasible. The network plan we're working on in our project follows the law and rules in the network field, and that outlined compliance measures and recommendations are followed. Proper licensing and network security protocols will help minimize legal risks and ensure that the network is secure and efficient.

Legal compliance also extends to the use of software and hardware. All network devices and software tools are used according to licensing agreements. To ensure continued compliance, the licensing agreements records will be kept. Compliance ensures that sensitive data is protected while reducing the risk of unauthorized access.

Cloud services are also compliant with their data protection policies and data transfer regulations. We ensure that cloud service providers meet legal standards for data protection and have clear agreements.

The use of SolarWinds Network Performance Monitor is legally feasible. Since the tool may collect and analyze network data, it is crucial to ensure that any monitored information does not include sensitive personal data, thereby avoiding potential violations of privacy regulations. The network monitoring process is conducted transparently. This can help in early detection of unauthorized access or anomalies, aiding in compliance with network security regulations and minimizing risks.

The use of VPN UTM for secure remote access is feasible, but it is still necessary to comply with cybersecurity laws and regulations. The firewall supports advanced security features like intrusion prevention to block unauthorized access. Implementing a legally compliant and strong and steady firewall helps prevent unauthorized access and data breaches and protect sensitive information.

Then, using VLANs to segment the network for different labs is legally feasible. It provides legal advantages by ensuring a safe and secure environment. This is relevant for maintaining compliance with data privacy laws that require limiting data access to authorized users only. Segmentation can help meet regulatory standards for data security, allowing maintenance and troubleshooting without compromising other parts of the network. We ensure proper VLAN configuration and regular audits to verify compliance with data security protocols.

5.2 Technical Feasibility

In order to satisfy the Faculty of Computing's requirements for fast connectivity and dependable connections, the new Block N28B will implement Category 6A Ethernet cabling that connects to managed gigabit switches. This configuration will enable growth to accommodate the projected 15% increase in numbers amongst students and staff within the next four years especially in such demanding areas as labs and student lounges.

To monitor and consolidate network traffic, bandwidth and the overall health of devices we suggest SolarWinds Network Performance Monitor (NPM). This way, through the use of NPM for real-time monitoring the potential problems will immediately be identified, therefore preventing interruption of connectivity to students and staff.

Due to the large amount of data, the backup system will be both on-site and a cloud platform such as Google Cloud or AWS depending on what is cheaper at the time of purchase for backup purposes. This evenly divided solution enables local application of high-speed and quick backup of the local app with another copy for disaster management in the event of data loss, thus safeguarding important data.

Every lab is going to have a different usage of the network. The Cisco Network Lab expects that the student uses dedicated networking equipment and the Embedded Lab focuses on IoT devices and sensors. Each lab will be set up optimally using Virtual LANs (VLANs), in which each lab will be isolated, thus making it easier to sort our problems, or make changes to, without affecting other sections. Smart whiteboards, cameras and collaboration systems will be incorporated in a hybrid classroom to allow for the physical classroom to include an online component as well.

For the mobile connectivity purpose Wi-Fi 6 access point will be placed at strategic locations in the building especially the student inside the building. Whereas, Wi-Fi 6 with enhanced speed and device capacity provides better wireless access for today and future users for smooth learning applications.

5.3 Financial Feasibility

A total of RM 1.5 million has been set aside for this project. Numerous aspects, including the cost of equipment, the quantity of routers and access points, the length of network cabling, and the kinds of service subscriptions needed, must be carefully taken into account in order to guarantee financial viability. When creating the floor plan for N28B, these factors will aid in cost containment.

The majority of the equipment will be purchased from UTM Digital in order to keep costs down because they are well-versed in the market and can provide premium goods at affordable costs. By utilizing their experience, we can get affordable solutions without sacrificing quality.

When it comes to service subscriptions, the emphasis will be on choosing the most cost-effective packages that satisfy all project specifications. For example, when selecting a Wi-Fi package, priority will be given to the most cost-effective option that guarantees fast network and 5G capability. This strategy strikes a balance between performance and cost effectiveness.

By optimizing equipment procurement and service subscriptions, the project can maximize production value while remaining within the allocated budget.

6.0 References

[1]"What is a Hybrid Backup? Cloud vs. Hybrid Backups Explained," *Data Backup*. https://www.techtarget.com/searchdatabackup/definition/hybrid-backup

[2]SolarWinds, "Network Monitoring Software | SolarWinds," *Solarwinds.com*, 2016. https://www.solarwinds.com/network-performance-monitor

[3]"What is a next-generation firewall (NGFW)?," *Cloudflare.com*, 2024. https://www.cloudflare.com/learning/security/what-is-next-generation-firewall-ngfw/

[4]"The Best Antivirus Protection for 2020," *PCMAG*. https://www.pcmag.com/picks/the-best-antivirus-protection

[5]SOLARWINDS, "What Is VLAN (Virtual LAN)? - IT Glossary | SolarWinds," *www.solarwinds.com*, 2024. https://www.solarwinds.com/resources/it-glossary/vlan

[6]T. Slattery and J. Burke, "What is a VLAN (Virtual LAN)?," *TechTarget*, May 2021. https://www.techtarget.com/searchnetworking/definition/virtual-LAN

[7]"What is Remote Monitoring and Diagnostics?," *www.clickmaint.com*. https://www.clickmaint.com/glossary/remote-monitoring-and-diagnostics

[8]i3-Technologies. (2023, May 8). *Welcome to the Hybrid Classroom - i3-Technologies*. Www.i3-Technologies.com. https://www.i3-technologies.com/en/blog/stories/education/welcome-to-the-hybrid-classroom/

[9]Legal. (2024). *Legal Feasibility and Financial Feasibility Analysis: How to Evaluate the Compliance and Regulatory Issues of Your Project - FasterCapital*. FasterCapital. https://fastercapital.com/content/Legal-Feasibility-and-Financial-Feasibility-Analysis--How-to-Evaluate-the-Compliance-and-Regulatory-Issues-of-Your-Project.html

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[11]Rutherford, J. (2023, August 16). *The Design of Hybrid Teaching Environments: 10 Questions Answered.* Er.educause.edu. https://er.educause.edu/articles/2023/8/the-design-of-hybrid-teaching-environments-10-questions-answered

[12]Writer, S. (2020, February 14). *Video Conferencing Setup Requirements: Your Checklist for Hardware and Software*. Lifesize. https://www.lifesize.com/blog/video-conferencing-requirements/

7.0 Appendix

7.1 Meeting Minute for Task 1

DATE/TIME	10 Oct 2024 3pm
LOCATION	KTDI DSR
AGENDA	Group Name & Task Distribution
Meeting MC	Chew Chuan Kai

ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
Chew Chuan Kai A23CS0062	1500	
Tan Min Xuan A23CS5030	1450	
ABID HUMAYRAA BINTI HARDISURA A23CS0032	1505	

MINUTES			
NO.	ITEM DISCUSSED	IDEA/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Find group name	- Chew Chuan Kai suggested Network Nexus - Tan Min Xuan suggested SignalBridge - Abid Humayraa suggested CommSphere	
2	Confirmation of group name	"Network Nexus" is our group name that achieved by every member after voting	
3	Software to use	We decided using "SmartDraw" , "Draw.io" and "floorplancreator.net" to design our floor plan	
4	Floor design	1. Chew Chuan Kai – Ground Floor's Floor Plan & Meeting Minute 2. Tan Min Xuan – First Floor's Floor Plan & Directory for floor plan 3. Abid Humayraa – Design Equipment needed in 4 labs, video conferencing room, hybrid classroom and student lounge.	
5	Meeting ended	1600	

7.2 Meeting Minute For Task 2

<table>
<tr><td>DATE/TIME</td><td colspan="2">23 Oct 2024 1.30pm</td></tr>
<tr><td>LOCATION</td><td colspan="2">Google Meet</td></tr>
<tr><td>AGENDA</td><td colspan="2">Task Distribution</td></tr>
<tr><td>Meeting MC</td><td colspan="2">Tan Min Xuan</td></tr>
<tr><td colspan="3">ATTENDANCE</td></tr>
<tr><td>NAME</td><td>TIME</td><td>REASON FOR ABSENCE</td></tr>
<tr><td>Chew Chuan Kai A23CS0062</td><td>1330</td><td></td></tr>
<tr><td>Tan Min Xuan A23CS5030</td><td>1330</td><td></td></tr>
<tr><td>Abid Humayraa Binti Hardisura A23CS0032</td><td>1345</td><td></td></tr>
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<table>
<tr><td colspan="4">MINUTES</td></tr>
<tr><td>NO.</td><td>ITEM DISCUSSED</td><td>IDEA/SUGGESTIONS AND PERSON GIVING IT</td><td>PERSON IN CHARGE & DATE</td></tr>
<tr><td>1</td><td>Task Distribution</td><td>Tan Min Xuan suggested assigning task equally to team members by one small topic and one feasibility.</td><td></td></tr>
<tr><td>2</td><td>Choosing Task</td><td>Chew Chuan Kai decided to take the service and financial feasibility as his task. Abid Humayraa decided to take the network and legal feasibility as her task. Lastly, Tan Min Xuan decided to take the faculty and technical feasibility as the task.</td><td></td></tr>
<tr><td>3</td><td>Final Task Distribution</td><td>Chew Chuan Kai – Service and Financial Feasibility
Tan Min Xuan – Faculty, Technical Feasibility and Meeting Minute
Abid Humayraa – Network and Legal Feasibility</td><td></td></tr>
<tr><td>4</td><td>Meeting ended</td><td>1430</td><td></td></tr>
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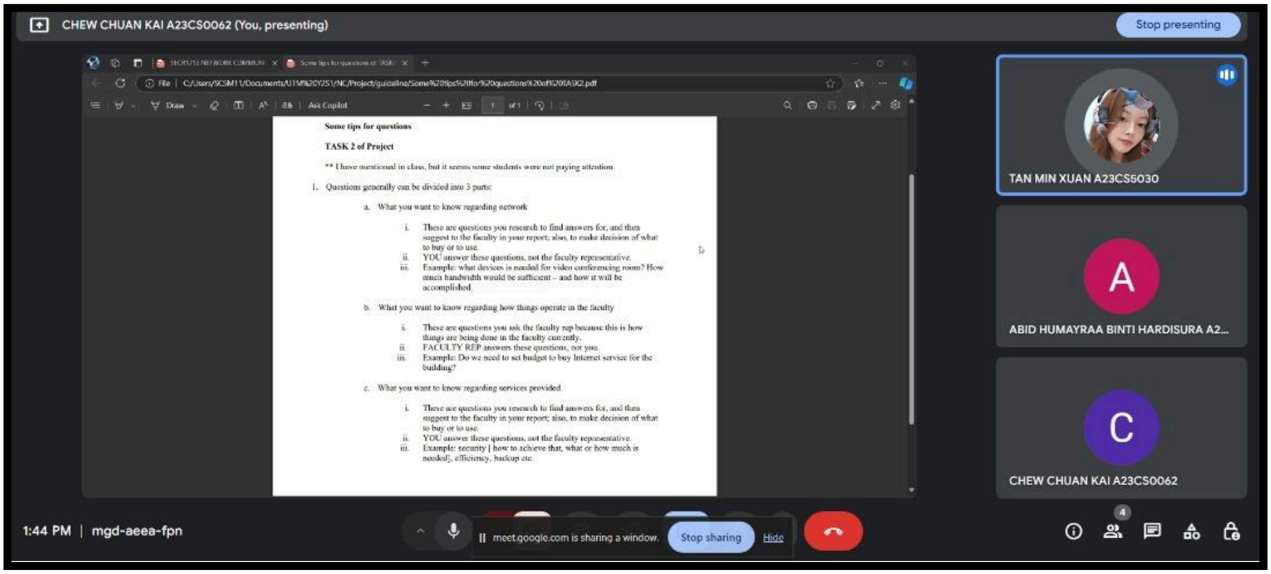

Figure above as an evidence of the google meet discussion for Task 2